

The reverse-Holder kernel hypothesis holds at every critical scale

FA/Banach run packet for arXiv:2602.03952

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Source question and intended conventions

Section 6 of P. Portal, *Using wave packet decompositions to construct function spaces*, arXiv:2602.03952, considers

$$H = -L = -\Delta + V \quad \text{on } \mathbb{R}^d, \quad d > 2,$$

where $V \geq 0$ belongs to RH_q for some $q > d/2$. It asks for the class of such potentials for which the kernel hypothesis (6.3) holds and suggests that the answer should be all of RH_q . We prove that suggestion.

We use the standard critical radius

$$\rho(x) = \sup \left\{ r > 0 : r^{2-d} \int_{B(x,r)} V \leq 1 \right\}. \quad (1)$$

The source prints the word min in this definition, but its examples $V = 1$ and $V(x) = |x|^2$, as well as the cited literature, force the standard supremum in (1). We also write the source multiplier as

$$T_\sigma = \psi(\sigma\sqrt{H}).$$

This is the operator meant by its displayed wave formula and its finite-speed scaling, although it is denoted there by $\psi(\sigma^2 L)$.

The source takes $\varphi \in C_c^\infty((0, \infty))$ and defines the even Schwartz function ψ by $\widehat{\psi}(\xi) = \varphi(|\xi|)$. Put $R_\psi = \sup\{|s| : s \in \text{supp } \widehat{\psi}\}$.

Full result

Theorem 1. *For every $A > 0$, T_σ has an integral kernel K_σ satisfying, for almost every $x, y \in \mathbb{R}^d$ and every $\sigma > 0$,*

$$|K_\sigma(x, y)| \leq C_A \sigma^{-d} \mathbf{1}_{\{|x-y| \leq R_\psi \sigma\}} \left(1 + \frac{\sigma}{\rho(x)}\right)^{-A} \left(1 + \frac{\sigma}{\rho(y)}\right)^{-A}. \quad (2)$$

Consequently the kernel hypothesis (6.3) of arXiv:2602.03952 holds for every $V \in RH_q$, $q > d/2$. In fact, the estimate holds at all scales, not only when $\sigma \geq \rho(x_j)$, and the boundedness of ρ is not needed for this kernel conclusion.

The proof uses one standard input. If $p_t(x, y)$ is the heat kernel of e^{-tH} , then for every $N > 0$,

$$0 \leq p_t(x, y) \leq C_N t^{-d/2} e^{-|x-y|^2/(ct)} \left(1 + \frac{\sqrt{t}}{\rho(x)} + \frac{\sqrt{t}}{\rho(y)}\right)^{-N}. \quad (3)$$

This is Theorem 4.10 of J. Dziubanski and J. Zienkiewicz, *H^p spaces associated with Schrodinger operators with potentials from reverse Holder classes*, Colloq. Math. 98 (2003), 5–38, in their notation $m = 1/\rho$. It is also restated as Proposition 29 in the included 2021 supporting paper.

Weighted resolvent columns

Lemma 1. *Given $A > 0$, there is an integer m such that the kernel a_σ of $A_\sigma = (I + \sigma^2 H)^{-m}$ satisfies*

$$\|a_\sigma(\cdot, y)\|_2 \leq C_A \sigma^{-d/2} \left(1 + \frac{\sigma}{\rho(y)}\right)^{-A}. \quad (4)$$

The same estimate holds for the row $a_\sigma(x, \cdot)$, with y replaced by x .

Proof. Choose m so that $\gamma = 2m - d/2 > A$. Positivity, self-adjointness, and the semigroup formula give

$$\begin{aligned} \|a_\sigma(\cdot, y)\|_2^2 &= (I + \sigma^2 H)^{-2m}(y, y) \\ &= \frac{1}{\Gamma(2m)} \int_0^\infty s^{2m-1} e^{-s} p_{s\sigma^2}(y, y) ds. \end{aligned}$$

Apply (3) on the diagonal with an exponent $N > 2\gamma$. If $a = \sigma/\rho(y)$, the right-hand side is at most

$$C \sigma^{-d} \int_0^\infty s^{\gamma-1} e^{-s} (1 + a\sqrt{s})^{-N} ds. \quad (5)$$

For $a \leq 1$, the integral is bounded. For $a > 1$, split it at a^{-2} and 1. The three pieces are respectively bounded by

$$C a^{-2\gamma}, \quad C a^{-2\gamma}, \quad C a^{-N}.$$

Since $\gamma > A$ and $N > 2\gamma$, (5) is at most $C_A \sigma^{-d} (1 + a)^{-2A}$. Taking square roots proves (4). Symmetry gives the row estimate. $\square$

Resolvent sandwich and finite propagation

Proof of Theorem 1. Fix $A > 0$, choose m from Lemma 1, and define

$$b(u) = (1 + u^2)^{2m} \psi(u), \quad B_\sigma = b(\sigma\sqrt{H}).$$

The function b is bounded because ψ is Schwartz. The spectral theorem gives the exact factorization

$$T_\sigma = A_\sigma B_\sigma A_\sigma, \quad \|B_\sigma\|_{2 \rightarrow 2} \leq \|b\|_\infty. \quad (6)$$

Thus (6) has the measurable kernel

$$K_\sigma(x, y) = \langle B_\sigma a_\sigma(\cdot, y), a_\sigma(\cdot, x) \rangle_{L^2},$$

up to the immaterial choice of complex conjugation convention. By Cauchy–Schwarz, (4), and its row version,

$$|K_\sigma(x, y)| \leq C_A \sigma^{-d} \left(1 + \frac{\sigma}{\rho(x)}\right)^{-A} \left(1 + \frac{\sigma}{\rho(y)}\right)^{-A}. \quad (7)$$

It remains only to recover the sharp spatial localization. Since ψ and $\widehat{\psi}$ are even,

$$T_\sigma = (2\pi)^{-1/2} \int_{-R_\psi}^{R_\psi} \widehat{\psi}(t) \cos(t\sigma\sqrt{H}) dt.$$

The wave equation for $H = -\Delta + V$ has propagation speed one; the zeroth-order potential does not enlarge supports. Hence the kernel of T_σ vanishes when $|x - y| > R_\psi \sigma$. Combining this with (7) proves (2).

Finally, the standard critical-radius comparability theorem gives $\rho(y) \simeq \rho(x_j)$ whenever $y \in Q_j$, because Q_j is a critical cube. Choose any $A > 1$, and set

$$g(r) = C_A \mathbf{1}_{(0, R_\psi]}(r), \quad h(s) = (1 + s)^{-A}.$$

Both functions belong to $L^1((0, \infty))$. Dropping the x -weight in (2) and replacing $\rho(y)$ by the comparable $\rho(x_j)$ yields, simultaneously for all j, k ,

$$|\mathbf{1}_{Q_k}(x) K_\sigma(x, y) \mathbf{1}_{Q_j}(y)| \leq \sigma^{-d} g\left(\frac{|x - y|}{\sigma}\right) h\left(\frac{\sigma}{\rho(x_j)}\right).$$

This is exactly (6.3), under the source's intended notation. □

Checks, strengthening, and novelty

The direct heat-semigroup route is not enough by itself: a general Schwartz wave multiplier need not be a Laplace transform. A one-sided resolvent factorization still leaves a pointwise-kernel gap. The two-sided sandwich (6) is what converts the known diagonal heat estimate into a pointwise bound by pairing two L^2 columns. Arbitrarily high resolvent order gives arbitrarily high critical-radius decay.

The proof establishes more than requested: symmetric decay in both critical radii, exact finite propagation, validity for every $\sigma > 0$, and no use of the source's global assumption $\rho \in L^\infty$ for the kernel estimate itself. A bounded search on 13 August 2026 for the source title, arXiv id, the named upcoming collaborators, and combinations of wave packets, reverse Holder potentials, and critical radii found no posted follow-up or prior statement of this resolution.

References

- P. Portal, *Using wave packet decompositions to construct function spaces: a user guide*, arXiv:2602.03952 (2026).
- J. Dziubanski and J. Zienkiewicz, *H^p spaces associated with Schrodinger operators with potentials from reverse Holder classes*, Colloq. Math. 98 (2003), 5–38.
- N. N. Trong, L. X. Truong, and T. D. Do, *Boundedness of second-order Riesz transforms on weighted Hardy and BMO spaces associated with Schrodinger operators*, C. R. Math. 359 (2021), 687–717.
- C. Remling, *Finite propagation speed and kernel estimates for Schrodinger operators*, Proc. Amer. Math. Soc. 135 (2007), 3329–3340.

Human-review recommendation. Review as a short full proof. The key checks are the exponent split in Lemma 1, the exact spectral factorization (6), and the interpretation of the source notation recorded at the start.